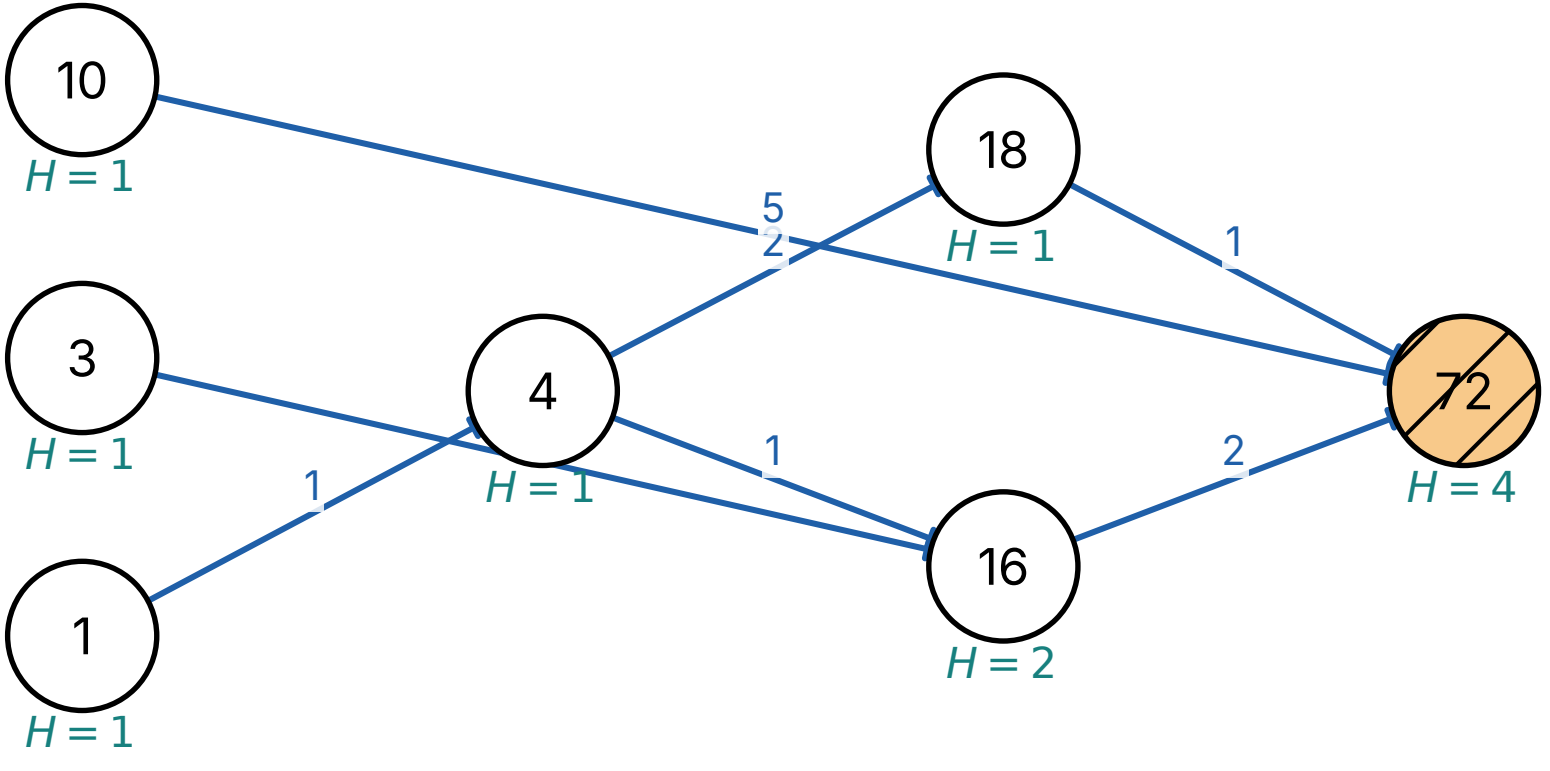$$H(t) = \sum_{d \geq 2,\ d^2 \mid t} H\left(\frac{t(d-1)}{d^2}\right);\ H = 1 \text{ for squarefree } t$$